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Key Points:

- We modeled the geomagnetic reversal frequency using adaptive-bandwidth kernel density estimation and found several troughs
- Recently discovered reversals at ∼31 Ma added to geomagnetic polarity time scale suppressed one trough around 32 Ma
- We concluded that the troughs in geomagnetic reversal frequency indicate missing geomagnetic reversals

Supporting Information:

Supporting Information may be found in the online version of this article.

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Evidence for Missing Geomagnetic Reversals From Geomagnetic Reversal Frequency Model Using Adaptive Kernel Density Estimation

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Abstract The existence of missing geomagnetic reversals has been proposed, with potential for new magnetostratigraphic age controls. We estimate geomagnetic reversal frequency from 0 to 155 Ma using adaptive-bandwidth kernel density estimation (AKDE) to evaluate data sparseness and to assess how reversal frequency changes when recently identified geomagnetic reversals are incorporated into the geomagnetic polarity time scale (GPTS) data set. AKDE is a two-stage procedure that uses an initial density estimator based on an initial (pilot) bandwidth. We found that the pilot bandwidth determined using cross-validation is stable with respect to data set length. The AKDE results obtained based on the cross-validated pilot bandwidth reveal four troughs after the Cretaceous Normal Superchron, spaced 13.5–15.0 Myr apart and corresponding to relatively long chrons (>0.8 Myr). One trough near 32 Ma becomes less distinct after the four recently identified reversals are added to the data set. This sensitivity suggests that troughs in the frequency curve may indicate missing geomagnetic reversals.

Plain Language Summary Earth's magnetic field has flipped many times in the past. Geologists use the timing of these flips, summarized in the geomagnetic polarity time scale (GPTS), to date rocks and sediments, but some flips may be missing from the standard timeline. We employed a statistical method of adaptive-bandwidth kernel density estimation to estimate the frequency of flips that occurred between approximately 155 million years ago and the present, and we inquired about the impact on this estimate when four newly recognized flips around 31 million years ago are incorporated into the GPTS timeline. Our analysis reveals fine-scale features of the geomagnetic reversal frequency over time and four intervals with unusually few flips after the end of a long, stable period in the Cretaceous. These low-activity intervals are separated by about 12–15.5 million years and align with unusually long periods of constant magnetic polarity. When the four new flips are included, the low-activity interval near 32 million years ago becomes less distinct. This pattern suggests that low-activity intervals may mark places where flips are still missing from the record.

1. Introduction

The Geomagnetic Polarity Time Scale (GPTS) is a key reference for determining the absolute ages of sediments, volcanic sequences, and seafloor magnetic isochrons. On timescales of tens of millions of years, geomagnetic reversals have occurred frequently (e.g., the Jurassic hyperactivity period; Kulakov et al., 2019) and infrequently (e.g., the Cretaceous Normal Superchron, CNS; Yoshimura, 2022). Because the timescale of changes in reversal frequency is much longer than the convective overturning time in the outer core, reversal frequency is thought to be controlled by heat flux across the core–mantle boundary (CMB) (e.g., Biggin et al., 2012; Larson &

Olson, 1991; McFadden & Merrill, 1984). Geodynamo simulations indicate that total CMB heat flow, as well as the pattern of heat-flux heterogeneity, influences reversal frequency (Amit & Olson, 2015; Aubert et al., 2025; Driscoll & Olson, 2011; Glatzmaier et al., 1999; Olson & Amit, 2015; Olson et al., 2010). Thus, geomagnetic reversal frequency serves as an important reconstructable proxy for estimating the history of CMB heat flux.

Several studies have suggested that short geomagnetic polarity intervals (i.e., some reversals) are missing from the GPTS, based on statistical analyses of chron lengths (Buffett & Avery, 2019; Marzocchi, 1997; McFadden, 1984). Missing reversals influence both age determination and our understanding of the geodynamo. Recently, four previously unrecognized polarity intervals were identified through high-resolution paleomagnetic and $^{40}Ar/^{39}Ar$ geochronology of Oligocene Ethiopian flood basalts (Ahn et al., 2021; Yoshimura et al., 2023). These intervals, known as the Lima-Limo reversals, raise the question of how their addition changes reversal-frequency models. Examining the influence of the Lima-Limo reversals can improve constraints on how temporal variations in reversal frequency relate to missing geomagnetic reversals.

Unfortunately, geomagnetic reversal frequency cannot be directly recorded in geological materials. Therefore, it must be modeled from one-dimensional GPTS age data. Several studies have modeled reversal frequency (e.g., Carbone et al., 2020; Constable, 2000; Gallet & Hulot, 1997; Hounslow et al., 2018; McFadden & Merrill, 1997; Melott et al., 2018; Sprain et al., 2016) and showed the common broad features on timescales of tens of millions of years. Some studies report uneven patterns and quasi-periodicities of ∼15 Myr in reversal frequency (Marzocchi & Mulargia, 1992; Mazaud et al., 1983; Mazaud & Laj, 1991; Stothers, 1986), whereas others argue that such structures do not exist (e.g., Constable, 2000).

Kernel density estimation (KDE) expresses a distribution as a sum of kernel functions (Wand & Jones, 1995). This approach is suitable for modeling reversal frequency because geomagnetic reversals can be treated as samples generated by a stochastic point process. Adaptive-bandwidth KDE (AKDE) is a two-stage procedure that uses an initial density estimator based on an initial bandwidth referred to as the pilot bandwidth. Constable (2000) modeled smooth reversal frequency from 160 Ma to the present using AKDE with the pilot bandwidth calculated using Silverman's rule of thumb (see Section 2.2), which is for the hybrid GPTS data set of Harland (1990) and Cande and Kent (1995).

In this study, we modeled geomagnetic reversal frequency using the AKDE. Because the true density underlying the KDE is unknown, an absolute ground truth cannot be obtained, and KDE results can vary depending on the assumptions. To address this, we aim to refine the standard AKDE so that it more closely approximates the true density by performing computational experiments in which KDE parameters are systematically varied. In particular, the parameters include the length of the data set, the pilot bandwidth, and the sensitivity parameter (α) of the adaptive bandwidth factor. By exploring reasonable choices for these parameters, this study seeks to estimate geomagnetic reversal frequency using the AKDE and to obtain a finer structure of reversal frequency curve than that reported by Constable (2000). The mathematical definitions and details of the pilot bandwidth and the sensitivity parameter are given in Section 2.2.

2. Data and Methods

2.1. Age Data of Geomagnetic Polarity Reversals From 155 to 0 Ma

We used data from the latest GPTS, edited in 2020 (Ogg, 2020), which does not include geomagnetic excursions. The GPTS has been continuously refined and is the basis for calculating the geomagnetic reversal frequency. The age model for the C-sequence (C34–C1) prior to GPTS2012 was based on a function of the distance from seafloor spreading centers (Cande & Kent, 1992, CK92); the GPTS was constructed by synthesizing C-sequence marine magnetic anomalies (MMA) in the Atlantic, Pacific, and Indian Oceans. CK92 was the first fully synthesized GPTS for the C-sequence with an age model of the geomagnetic reversal boundaries from the radiometric ages. CK92 was revised to CK95 (Cande & Kent, 1995) using the latest age of the K-Pg boundary at that time. In the C-sequence age model of the geologic time scale 2004 (GTS2004), astronomical tuning tied to the present was applied since the late Neogene (Lourens et al., 2004). In GTS2012, the astronomical tuning was also applied to the Paleogene and late Cretaceous (Gradstein et al., 2012). Finally, in GTS2020, the entire C-sequence from the late Cretaceous to the Cenozoic was astronomically tuned without only 2-Myr interval of C6AA–C6A (Ogg, 2020). The Mesozoic age model, the M-sequence, is expressed as a function of distance from Chron M0r using Pacific magnetic anomaly profiles (e.g., Channell et al., 1995, CENT95). In contrast to the C-sequence modified by astronomical tuning along an almost full sequence, in the M-sequence, the age model is constrained by a partial astronomical tuning and a few radiometric ages (Ogg, 2020). Furthermore, the onset of the Aptian (the onset of Chron M0r) was shifted from approximately 126 Ma to 121 Ma in GTS2020 (Ogg, 2020). We analyze the period younger than Chron M27r, which included only well-established Chrons.

At the boundaries, edge effects occur due to the temporal limitations of the data set. Therefore, the GPTS2020 data set has been extended by reflecting it at both boundaries from $\{0, t_1, t_2, \ldots, t_N\}$ to $\{-t_N, \ldots, -t_2, -t_1, 0, t_1, t_2, \ldots, t_{N-1}, t_N, 2t_N - t_{N-1}, 2t_N - t_{N-2}, \ldots, 2t_N - t_1\}$. As an exception, when the end of the data includes the Cretaceous Normal Superchron (CNS), only the opposite boundary to the CNS is reflected. Following Constable (2000), we performed separate calculations before and after the CNS (0–121 Ma and 83–155 Ma). The statistical basis of this method can be found in Jones (1993).

The Lima-Limo reversals were initially suggested by the high-resolution paleomagnetic study (Ahn et al., 2021) and defined by the $^{40}Ar/^{39}Ar$ geochronological study (Yoshimura et al., 2023). We assumed that the Lima-Limo reversals are evenly distributed over the earlier half of Chron C12n, with ages of 30.82, 30.86, 30.90, and 30.94 Ma. This is a tentative assignment because the direct ages of the Lima-Limo reversals are unknown.

2.2. Kernel Density Estimation

KDE represents a distribution as a sum of kernel functions and is suitable for representing the geomagnetic reversal frequency. If geomagnetic reversals are regarded as deterministic processes (i.e., the reversal timing is predictable), then a delta function can be placed at the time of each reversal, and the resulting probability density function will be zero anywhere else, but such predictions are not possible because reversal timing is random (Constable, 2000). A plausible model for the occurrence of geomagnetic reversals is a renewal process, driven by random instabilities in fluid motion within the outer core. If the geomagnetic reversals frequency is affected by mantle convection, the frequency can be considered as a time-varying non-stationary stochastic point process on the timescales of tens of millions of years.

In this study, we do not specify the model explicitly; we instead regard the observed geomagnetic reversals as a sample of size N from some stochastic point process and estimate the geomagnetic reversal frequency non-parametrically using KDE (Wand & Jones, 1995). We adopted the Gaussian kernel, which is widely used as a kernel function, and computed the geomagnetic reversal frequency variations using the akj functions in the R package quantreg (Koenker, 2010) for AKDE.

Here we summarize the procedure of the AKDE used in this paper. The AKDE begins with the initial density estimator (referred to as pilot density) with a bandwidth h (referred to as pilot bandwidth) defined as follows:

$$f(t; h) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{t - t_i}{h}\right),$$

where K is a function such that $\int K(t)\, dt = 1$ and is called the kernel. This equation is the average of the kernel function values at $(t - t_i)/t$. The pilot bandwidth parameter $h > 0$ controls the time resolution of the event occurrence. In this study, we used the following Gaussian function of $N(0, 1)$ as kernel function:

$$K(t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right).$$

Two methods are available for selecting the bandwidth parameter h. The first method is based on the minimizing the Mean Integrated Squared Error (MISE):

$$\mathrm{MISE}\{f(t; h)\} = E \int \{f(t; h) - f(t)\}^2 dt = E \int f(t; h)^2 dt - 2E \int f(t; h) f(t)\, dt + \int f(t)^2 dt.$$

The MISE involves a true probability density function which is unknown and cannot be evaluated directly. Instead, we use Least-Squares Cross-Validation (LSCV) as the criterion for the minimization. The optimal h is obtained as the minimizer of $\mathrm{LSCV}(h)$ defined below:

$$\mathrm{LSCV}(h) = \int f(t;h)^2 dt - \frac{2}{n}\sum_{i=1}^{n} f_{-i}(t_i;h),$$

where

$$f_{-i}(t;h) = \frac{1}{(n-1)\,h}\sum_{j\neq i}^{n} K\left(\frac{t-t_j}{h}\right).$$

The LSCV is an approximately unbiased estimator of (the first two terms of) the MISE based on the observed data. For the derivation, see Silverman (1986), Section 3.4.3). To compute $\mathrm{LSCV}(h)$, we use the R function h. ucv in the R package kedd (Guidoum & Giné-Vázquez, 2024).

Note that both the KDE $f(t;h)$ and the leave-one-out KDE $f_{-i}(t;h)$ are normalized such that $\int f(t;h)\,dt = \int f_{-i}(t;h)\,dt = 1$ and therefore have physical dimension of the inverse time (Myr^{-1}). The quantity $\mathrm{LSCV}(h)$ has the same physical dimension.

The second method is Silverman's "rule of thumb" (Silverman, 1986):

$$h = 0.9 \times \min\left(\hat{\sigma}, \frac{\mathrm{IQR}}{1.34}\right) \times n^{-1/5}$$

where $\hat{\sigma}$ is a standard deviation of the data set.

AKDE adjusts the pilot bandwidth by locally stretching or shrinking it according to the sparsity of the data. The sparseness is determined based on the geometric mean of the initially calculated KDE, and the adaptive bandwidth factor λ_i is calculated according to this sparseness using the following relationship (Silverman, 1986 (5.7)):

$$\lambda_i = \left\{\frac{f(t_i;h)}{m}\right\}^{-\alpha},$$

where α is the sensitivity parameter, which takes the values 0 to 1, and m is the geometric mean of $f(t_i;h)$:

$$\log m = \frac{1}{n}\sum_{i=1} \log f(t_i;h).$$

Then, the AKDE is defined as

$$\hat{f}(t;h) = \frac{1}{nh}\sum_{i=1}^{n}\frac{1}{\lambda_i}K\left(\frac{t-t_i}{\lambda_i h}\right)$$

Using the AKDE defined above, we conducted four types of data analysis (1)–(4), the results of which are summarized in Sections 3.1–3.4, respectively. In each analysis, we examined the effects of data set length, pilot bandwidth, and the sensitivity parameter α on the AKDE.

1. The pilot bandwidth was calculated using Silverman's "rule of thumb." We tested whether the shape of the AKDE using the pilot bandwidth depends on the length of the GPTS2020 data set. Adaptive bandwidths corresponding to data sets of different lengths were also computed. Sensitivity parameter α was varied to 0.1, 0.5, and 0.9, and KDEs were calculated for each value.
2. For the same procedure as (1), the pilot bandwidth was determined by cross-validation, and the AKDE and adaptive bandwidths were computed for data sets of different lengths. Sensitivity parameter α was varied in the same way as (1).
3. We calculated the AKDE median and interquartile range (IQR) using the 1000 AKDEs based on the pilot bandwidth determined in (2). 1000 AKDEs were calculated from 1000 bootstrap GPTS2020 samples.

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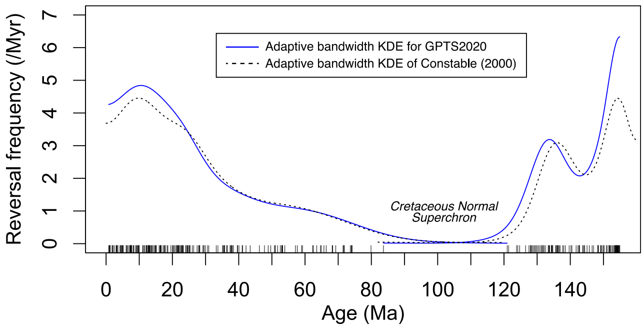

Figure 1. Geomagnetic reversal frequency variations from 155 to 0 Ma based on the GPTS2020 modeled using adaptive-bandwidth kernel density estimation whose pilot bandwidth is calculated using Silverman's rule (Blue solid curves). The timing of geomagnetic reversals is shown as an event plot at the bottom of the reversal frequency plot. Black dashed curves are the models by Constable (2000) based on the hybrid geomagnetic polarity time scale data set of Harland (1990) and Cande and Kent (1995).

4. For the GPTS2020 data set that includes the Lima-Limo reversals, we performed the same procedure as in (3) and compared the results with those obtained in (3).

3. Results

3.1. AKDE for GPTS2020 With the Pilot Bandwidth Calculated Using Silverman's Rule

When Silverman's rule was used to calculate the pilot bandwidth, the AKDE corresponding to the 0–121 Ma data set is overall smooth (Figure 1; Figure S1 in Supporting Information S1). From ~155 Ma to ~121 Ma, reversal frequency decreased toward the CNS, forming a single trough. It then increased from ~83 Ma to ~10 Ma after the CNS and decreased again from ~10 Ma to the present. The shape of the AKDE matches the results of Constable (2000) (black dashed line in Figure 1 and Figure S1; Table S1 in Supporting Information S1), demonstrating the reproducibility of the AKDE method. As the GPTS data sets become shorter (0–83 Ma, 0–70 Ma, 0–60 Ma, and 0–50 Ma), fine-scale structure in the AKDE becomes more apparent (Figure S1 in Supporting Information S1). The overall characteristics remain largely unchanged when α is varied; however, for the 0–50 Ma data set, fine-scale structure becomes more distinct with $\alpha = 0.9$ during high-density intervals (0–~25 Ma) and with $\alpha = 0.1$ during low-density intervals (~25–~50 Ma). The pilot bandwidths calculated using Silverman's rule for data sets of different lengths were 6.562, 6.501, 5.709, 4.904, and 3.990, respectively (Figure 2). The adaptive bandwidth also became narrower as the data set length decreases in general (Figure S1 in Supporting Information S1).

3.2. AKDE for GPTS2020 With the Pilot Bandwidth Determined Using Cross-Validation

Next, when the pilot bandwidth was determined by cross-validation, fine-scale structures also appeared in the AKDEs for all data sets: 0–121 Ma, 0–83 Ma, 0–70 Ma, 0–60 Ma, and 0–50 Ma (Figure S2 in Supporting Information S1). Except for the 0–121 Ma data set, the adaptive bandwidths ranged approximately from 0 to 4. Here, we describe the ages of the peaks and troughs using an AKDE with $\alpha = 0.5$ as a representative example. Four troughs are observed at 17.9, 31.8, 45.4, and 60.3 Ma, while five peaks occur at 10.1 Ma, 22.3 Ma, 37.1, 52.2, and 67.4. For the 83–155 Ma data set, two troughs were identified in the AKDE. Even when the GPTS data sets were shortened to 0–83 Ma, 0–70 Ma, 0–60 Ma, and 0–50 Ma, the pilot bandwidths determined by cross-validation remained stable around three, with values of 3.087, 3.097, 2.998, 2.929, and 3.128, respectively (Figure 2 and Figure S3 in Supporting Information S1). Consequently, regardless of data set length, the adaptive bandwidths are also stable (Figure S2 in Supporting Information S1).

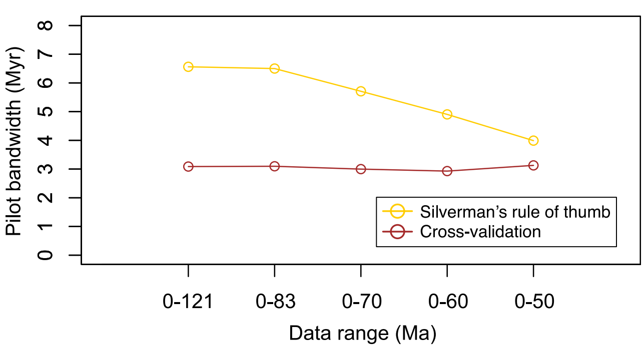

Figure 2. Relationship between data set range (i.e., length) and pilot bandwidth determined using Silverman's rule or cross-validation.

3.3. IQR Envelope Calculation of AKDE for GPTS2020

Using bootstrap resampling ($n = 1000$), we computed AKDEs and calculated IQR envelopes (Figure S4 in Supporting Information S1). The pilot bandwidths determined using cross-validation were used for the adaptive bandwidth because they are less affected by data set length. Since the overall pattern of troughs and peaks did not depend strongly on data set length, we computed AKDEs for the 0–121 Ma and 83–155 Ma data set. AKDEs with IQR envelopes were produced for α values of 0.1 (Figure 3a), 0.5 (Figure 3b), and 0.9 (Figure 3c). For the 0–121 Ma interval, the major peaks and troughs can be identified even considering the IQR envelopes. For the 83–155 Ma data set, finer-scale structures are also identifiable considering the IQR envelopes.

The sharpness of troughs (local minima) and peaks (local maxima) varies with α. When α is 0.1, the 45.4 and 60.3 Ma troughs are pronounced, whereas the 22.3 Ma peak is muted. Conversely, when α is 0.9, the 45.4 and 60.3 Ma troughs weaken, and the 22.3 Ma peak strengthens. When α is 0.5, the KDE represents an intermediate feature between $\alpha = 0.1$ and 0.9.

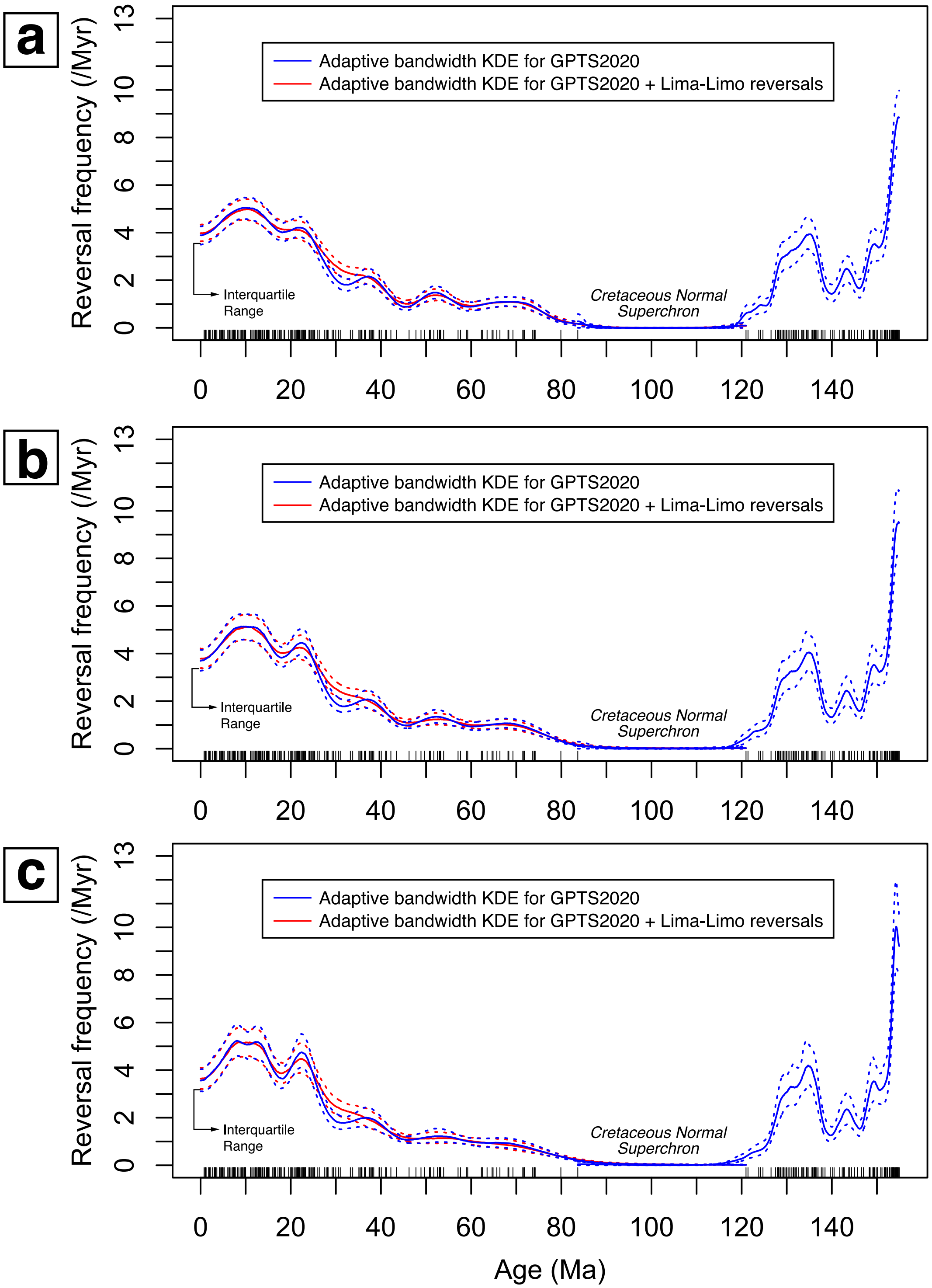

Figure 3. Adaptive-bandwidth kernel density estimation median with the cross-validated pilot bandwidth with IQR for GPTS2020 (blue lines) and GPTS2020 with the Lima-Limo reversals (red lines). The case of (a) $\alpha = 0.1$, (b) $\alpha = 0.5$, and (c) $\alpha = 0.9$.

3.4. AKDE With IQR for GPTS2020 Including the Lima-Limo Reversals

The geomagnetic reversal frequency curve for GPTS2020, including the Lima–Limo reversals modeled with the AKDE and IQR envelopes, shows features similar to those described in Sections 3.2 and 3.3. A notable difference is that the trough at 31.8 Ma becomes less distinct after adding the four reversals (red curves in Figure 3).

4. Discussion

4.1. Pilot Bandwidth Used for AKDE

The pilot bandwidth estimated using Silverman's rule decreases from approximately seven to four as the length of the GPTS data set shortens (Figure 2). As a result, the adaptive bandwidth also narrows (Figure S1 in Supporting Information S1). This likely occurs because, even as the time interval increases, the total number of data points does not increase substantially. Consequently, the increasing time interval of the data leads to larger estimates of the standard deviation $\hat{\sigma}$ and IQR of the average of data set (see the definition of Silverman's rule in Section 2.2). In contrast, the pilot bandwidth determined by cross-validation remains stable at around three, regardless of the data set length (Figure 2). This suggests that the cross-validated pilot bandwidth provides a more robust AKDE than that based on Silverman's rule. Our results show that using Silverman's rule smooths small-scale structures (Figure 1; Figure S1 in Supporting Information S1), such as the uneven patterns captured by the AKDE based on the cross-validated pilot bandwidth (Figure 3; Figure S2 in Supporting Information S1). Therefore, selecting an appropriate pilot bandwidth is essential when calculating the AKDE.

4.2. Peaks and Troughs in the AKDE

The AKDE based on the cross-validated pilot bandwidth resolves structures on the scale of a few million years, whereas the AKDE using Silverman's rule misses most of these small-scale structures but preserves broader, tens-of-millions-of-years trends. We identify troughs corresponding to longer polarity Chrons (>0.8 Myr) in GPTS2020 (after the CNS: Chrons C5Br, C6n, C12r, C20r, C24r, C25r, and C26r; before the CNS: Chrons M16n, M17r, and M22n.1n). The peaks in the AKDE appear to correspond to intervals with frequent geomagnetic reversals (event plot in Figures 1 and 3). Thus, the troughs represent periods of sparse reversals, while the peaks reflect periods of relatively frequent reversals. The IQR envelopes distinguish most peaks and troughs reasonably well. Although some features are only partly resolved within the envelopes, they retain their overall shape, indicating that the overall peak–trough structure is robust. However, we do not claim that all observed peaks and troughs represent true variations in reversal frequency, because troughs may simply reflect lack of data. In addition, although we tried to evaluate the troughs in the AKDE using synthetic data set generated from non-homogenous Poisson process, we concluded the non-homogenous Poisson process is limited for this type of validation (see Figures S5 and S6 in Supporting Information S1).

To evaluate quasi-periodicity, we calculated the time intervals between neighboring troughs and between neighboring peaks. Troughs at 17.9, 31.8, 45.4, and 60.3 Ma yield intervals of 13.9, 13.6, and 14.9 Myr. Peaks at 10.1, 22.3, 37.1, 52.2, and 67.4 Ma yield intervals of 12.2, 14.8, 15.1, and 15.2 Myr. The characteristic time intervals of 12–15.5 Myr agree with the previously proposed ∼15 Myr quasi-periodicity in geomagnetic reversal frequency (Marzocchi & Mulargia, 1992; Mazaud et al., 1983; Mazaud & Laj, 1991). They are also nearly consistent with one of the four quasi-periods (16 Myr) identified by Melott et al. (2018) from reversal frequencies over the past 375 Myr. In the present study, the newly obtained AKDE provides independent support for characteristic quasi-periodic intervals of 12–15.5 Myr. Mazaud and Laj (1991) argued that a monotonic random process is unlikely to generate such quasi-periodic behavior in reversal sequences. Following their perspective, our results suggest that the non-stationarity of geomagnetic reversal is also supported by the AKDE analysis. However, the cause of these quasi-periodic intervals remains unresolved.

4.3. Less Distinct Troughs of the Geomagnetic Reversal Frequency by Adding Newly Discovered Reversals and Evidence of Missing Geomagnetic Reversals

We examined how the Lima–Limo reversals influence the shape of our geomagnetic reversal frequency curve. When the Lima–Limo reversals were added to the GPTS2020 data set and the AKDE was calculated using the cross-validated pilot bandwidth, the trough near Chron C12r (at 31.8 Ma) became less distinct (red line in Figure 3), and the result almost differed from the AKDE for GPTS2020 without the Lima–Limo reversals at the

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IQR level. The Lima–Limo reversals occur within Chron C12n, adjacent to the relatively long Chron C12r (2.237 Myr, according to Ogg, 2020). These observations suggest that missing geomagnetic reversals may exist near the edges of long Chrons such as C12r. Buffett and Avery (2019) proposed that short polarity intervals (<~30 kyr), including geomagnetic excursions or precursor and rebound features linked to polarity transitions, are likely to be undetected in MMA. Combined with the discussion in Section 4.2, the missing reversals may occur at ages that correspond to troughs in the AKDE, in other words, near or within longer Chrons (>0.8 Myr). This idea is consistent with the observation that seven reversed-polarity events appear to be concealed within the CNS (e.g., Zhang et al., 2021). The troughs in the AKDE-based geomagnetic reversal frequency curve may therefore serve as targets for future high-resolution paleomagnetic studies seeking to identify missing reversals. If such reversals are discovered at ages corresponding to AKDE troughs, the troughs in the AKDE would become less distinct, producing a smoother and more gradual reversal frequency variations similar to the pattern reported by Constable (2000) (Figure 1). We consider such a smoothed reversal frequency curve to represent actual variations. Numerical geodynamo simulations have shown that geomagnetic reversal frequency is strongly modulated by the thermal boundary conditions at the CMB, particularly by the amplitude and spatial pattern of the CMB heat flux (e.g., Driscoll & Olson, 2011; Glatzmaier et al., 1999; Olson et al., 2010). Importantly, variations in the CMB heat flux are expected to occur on very long timescales, typically on the order of hundreds of millions to a billion years, reflecting the sluggish nature of mantle convection and true polar wander processes (e.g., Frasson et al., 2024). Such long-wavelength and temporally smooth variations in the CMB heat flux make it unlikely that sharp and short-lived troughs in geomagnetic reversal frequency represent primary geodynamo behavior, supporting the interpretation that these troughs may instead reflect incomplete detection of short polarity intervals.

In our AKDE-based geomagnetic reversal frequency model, the present appears to correspond to the most recent trough (Figure 3). In this context, the latest characteristic time interval of 17.9 Myr may have lengthened by ~3–~4.3 Myr relative to the 13.6–14.9 Myr interval of identified troughs. This would mean that the present interval, Chron C1n (Brunhes), may correspond to the most recent trough. Chron C1n may contain as many as 12 geomagnetic excursions (equivalent to 24 reversals if treated as full reversals) (Singer et al., 2014). If excursions tend to occur near troughs in the reversal frequency, the excursions within Chron C1n could represent missing geomagnetic reversals. However, we must consider the case that this pattern may be affected by boundary correction introduced through reflection of the data set (see Section 2.1). Considering the 13.6–14.9 Myr characteristic interval of neighboring troughs after the CNS, the period apart from the trough at 17.9 Ma with 13.6–14.9 Myr is between 3 and 4.3 Ma (Chron C2An.1n, Gauss to C3n.1n, Gilbert), which may correspond to the period possessing missing reversals. Recent studies suggest that excursions may not be clearly distinguishable from polarity reversals (Buffett, 2024; Constable & Morzfeld, 2025), implying that future analyses of reversal frequency should treat excursions as potential polarity reversal events. From this perspective, adding the Lima–Limo reversals to the GPTS2020 data set is reasonable, even though they have not yet been confirmed as geomagnetic excursions or polarity reversals. Such verification requires paleointensity measurements, but the lava flows associated with the Lima-Limo polarity interval do not exhibit strong paleointensity (Yoshimura et al., 2020).

5. Conclusion

The geomagnetic reversal frequency for GPTS2020 was modeled using the AKDE. When modeling the AKDE, cross-validation is more appropriate method than Silverman's rule to determine the pilot bandwidth. The four troughs with 13.5–15.0 Myr time intervals after the CNS represent robust features in our AKDE model calculated using a cross-validated pilot bandwidth, as they remain identifiable considering the IQR. After adding the Lima-Limo reversals, the trough near 32 Ma became less distinct. We concluded that troughs in the AKDE-based geomagnetic reversal frequency that correspond to intervals containing fewer reversals may indicate the existence of missing geomagnetic reversals during those periods.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The reversal frequency models can be downloaded from https://earthref.org/ERDA/2772/.

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